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RESEARCH ARTICLE

Real-Time Dynamic Gesture Recognition Method Based on Gaze Guidance

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ABSTRACT The application of gesture recognition technology in human-computer interaction fields is widespread. However, issues such as the size of the model parameter space and the occurrence of false positives in real-world interactive scenarios persist. This paper proposes a real-time dynamic gesture recognition method based on gaze guidance, which utilizes gaze tracking data to accurately segment hand skeletal sequences. Furthermore, skeletal data is pre-processed with multidimensional feature extraction. Finally, a lightweight multi-feature fusion recognition network is employed for the gesture recognition. Experiments on public datasets and simulated interaction scenarios demonstrate that the proposed method achieves higher recognition accuracy than mainstream methods, using only approximately 0.15M parameters and 3ms inference time. In particular, in the simulated 14-class and 28-class dynamic gesture recognition tasks, it obtains Levenshtein accuracies of 95.9% and 94.5%, respectively, which are approximately 20% higher than those of mainstream methods.

INDEX TERMS Hand gesture recognition, gaze guidance, mixed reality, deep learning network.

I. INTRODUCTION

Gesture recognition, a crucial modality in human-computer interaction, has witnessed a steady evolution in recent years and has been extensively deployed across a multitude of domains, particularly in the realms of virtual reality interaction [1], [2], robotic teleoperation [3], [4], and video gaming [5], [6].

In contrast to the extensive research conducted on gesture recognition methods based on RGB or depth images [7], [8], [9], [10], the investigation of gesture recognition based on skeleton data has emerged as a promising area of inquiry. This approach offers the advantage of greater adaptability to dynamic environments, while simultaneously achieving higher computational efficiency and recognition accuracy. This paper also addresses the topic of dynamic gesture recognition for skeleton data.

Skeleton-based dynamic gesture recognition approaches can be categorized into two main groups: traditional handcrafted feature-based methods and deep learning-based methods. The former employs motion features that have been meticulously designed and are compact, but lack efficiency and comprehensive data capture capabilities. The advent of deep learning technologies has led to the emergence of numerous deep learning-based dynamic gesture recognition methods, which demonstrate high recognition accuracy and strong generalization capabilities across various challenging datasets. However, the majority of these methods are computationally intensive and time-consuming, rendering them inadequate for real-time interactive scenarios where rapid user response is crucial. To address this issue, this paper proposes a lightweight multi-feature fusion recognition network. This network employs a minimal number of parameters while achieving recognition performance superior to mainstream methods across multiple datasets.

In practical human-computer interaction scenarios, such as mixed reality (MR), gesture data is continuously generated during interactions, making it challenging to define the temporal window for dynamic gestures. Recent studies have sought to address this issue; however, due to the complexity

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of the scenes and the presence of excessive noise, these approaches continue to exhibit high false trigger rates, failing to meet application requirements. To overcome this limitation, this paper proposes a real-time gaze-guided recognition trigger mechanism specifically designed for mixed reality interaction scenarios. This mechanism effectively addresses the temporal window segmentation challenge, significantly reduces the false trigger rate of gesture recognition algorithms, and enhances their overall usability. The primary contributions of this paper are as follows:

1) We introduce a novel lightweight multi-feature fusion recognition network designed to mitigate the high computational complexity prevalent in dynamic gesture recognition.

2) We develop a real-time gaze-guided recognition triggering mechanism specifically for mixed reality scenarios. This mechanism enables the natural and precise segmentation of dynamic gestures, markedly enhancing the usability of gesture recognition algorithms.

3) Leveraging HoloLens 2, we create a new dynamic hand gesture dataset, HoloGesture. This dataset offers more precise hand pose recognition and an expanded number of samples, providing a valuable resource for future research in the field.

The remainder of this paper is structured as follows: In Sect. II, we review the related work on isolated gesture recognition and real-time gesture recognition. In Sect. III, we provide a comprehensive description of the proposed model. In Sect. IV, we introduce the datasets used in this study, present the experimental results, and conduct a comparative analysis with state-of-the-art approaches. Finally, Sect. V is devoted to discussion and future perspectives.

II. RELATED WORK

A. ISOLATED GESTURE RECOGNITION

Isolated gesture recognition has been extensively studied. Among skeleton-based dynamic gesture recognition methods, they are primarily categorized into handcrafted feature-based approaches and deep learning-based approaches. Some early works utilized handcrafted features, including skeletal joint coordinates, joint geometric features, and motion features. For example, Yang and Tian [11] proposed a novel feature descriptor that models the dynamic information of each joint by leveraging temporal and spatial differences of skeletal joints. This is then used by a non-parametric Naive Bayes Nearest Neighbor (NBNN) classifier to recognize different actions. De Smedt et al. [12], [13] described hand direction using wrist and palm joints, hand shape using grouped joint coordinates, and hand motion using joint velocities, with a linear SVM classifier employed for gesture classification. While these traditional handcrafted feature-based methods are compact and efficient, they are unable to accurately recognize complex gestures.

In contrast, deep learning-based methods offer superior feature representation capabilities, capturing more abstract and high-level gesture features, and have achieved significant performance improvements in dynamic gesture recognition. Nunez et al. [20] employed a deep learning approach combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. Their experiments demonstrated that the CNN model architecture could extract relevant features from three-dimensional skeletal data, which the LSTM then dynamically recognized. Given that hand skeletal data form a graph structure, Graph Neural Networks (GNNs) are well-suited for modeling these data, effectively capturing the connectivity between joints [14]. Yan et al. [21] and Chen et al. [22] adopted Graph Convolutional Network (GCN) architectures to extract spatiotemporal features based on joint connections. However, the computational demands of graph convolutions exceed those of standard convolutions, resulting in longer processing times and making them unsuitable for real-time interactive scenarios where rapid user response is essential.

To address this issue, this paper proposes a lightweight multi-feature fusion recognition network. This network uses several layers of one-dimensional standard convolutions to extract and fuse preprocessed hand skeletal features. Despite having a minimal number of parameters, the network outperforms mainstream methods on multiple datasets.

B. REAL-TIME GESTURE RECOGNITION

Most dynamic gesture recognition algorithms are designed to handle pre-segmented datasets, where each sample segment represents an isolated dynamic gesture. Although these algorithms achieve high recognition accuracy with isolated gesture data, they face significant challenges in real-world human-computer interaction scenarios, such as mixed reality (MR). In these scenarios, gesture data is continuously generated during interactions, making it difficult to determine the start and end points of dynamic gestures [15].

Two primary approaches have been developed to address the issue of temporal window segmentation. The first approach simultaneously performs segmentation and recognition, while the second approach independently detects the boundaries of each isolated gesture to achieve segmentation. For the first approach, Dynamic Time Warping (DTW) is a widely used method for segmenting continuous gesture sequences [16]. This method extracts feature vectors from gesture sequences, calculates the DTW distance to template gestures, and uses the Viterbi algorithm to identify appropriate temporal windows. The second approach typically involves the analysis of user gesture patterns and the establishment of segmentation rules based on specific scenarios. For example, Liu et al. [17] analyzed hand height variations to set height thresholds for gesture segmentation. Similarly, Köpüklü et al. [17] and Xu et al. [32] detected the presence of hands in the camera view to determine temporal segmentation boundaries.

The first type of method is computationally complex and exhibits low recognition accuracy. The second type is scenario-specific and closely tied to user behavior patterns, resulting in poor generalization and unnatural interactions.

Consequently, there is a need for a more natural and efficient method for temporal window segmentation.

In response to this challenge, this paper proposes a real-time gaze-guided recognition triggering mechanism specifically designed for mixed reality interaction scenarios. By utilizing gaze-tracking data to accurately segment hand skeletal sequences, this mechanism effectively addresses the temporal window segmentation problem, significantly reduces the false trigger rate in gesture recognition algorithms, and enhances the overall usability of the system.

III. METHODOLOGY

A. LIGHTWEIGHT MULTI-FEATURE FUSION RECOGNITION NETWORK

In real-world gesture interaction scenarios, it is not only essential to achieve high accuracy in gesture recognition, but also to limit the number of parameters in the algorithm and to accelerate the recognition speed. This is so that the algorithm can be deployed in various terminals more efficiently, thereby improving the user's interaction experience. Therefore, this paper proposes a lightweight multi-feature fusion recognition network for pre-processed multi-dimensional hand skeletal feature data. This network ensures the recognition accuracy while reducing the inference time and the number of parameters of the model.

1) FEATURE PREPROCESSING

We mainly utilize hand keypoint coordinate features for dynamic gesture recognition. The hand keypoint coordinates obtained from HoloLens 2 are expressed in world coordinate space and are contingent upon the user's position and camera view. Consequently, the raw coordinate features necessitate pre-processing in order to derive more recognition-friendly features. The skeletal data is processed from two distinct perspectives: local gestures and global gestures. In the case of local gestures, relative distance and bending angle features are extracted. With regard to global gestures, translation change and rotation change features are obtained.

- **Relative distance feature**

The relative distance feature is employed to describe hand gestures such as pinch, open, etc. This feature is independent of the position of the hand and hand rotation, and only reflects the relative distance of the joint points of the hand. In this paper, we calculate the Euclidean distance between pairs of key points of the hand to obtain a symmetric matrix. In order to reduce data redundancy, we only use the lower triangular part of the matrix in addition to the diagonal elements as the relative distance feature.

Specifically, assume that the total number of frames of a dynamic gesture sequence is N and the number of hand joints is L. In the n-th frame, the world coordinates of the l-th joint are denoted as $P_l^n = (x, y, z)$, and all the coordinates of the keypoints are merged to obtain the joint dataset $S^n = \left\{P_0^n, P_1^n, \ldots, P_{L-1}^n\right\}$ in the n-th frame, and S^n of the relative distance feature $F_{local_dis}^n$ is calculated as follows:

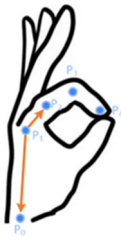


FIGURE 1. The bending angle feature.

$$F_{local_{dis}}^n = \begin{bmatrix} \left|\left|\overrightarrow{P_1^n, P_0^n}\right|\right| & & \\ \vdots & \ddots & \\ \left|\left|\overrightarrow{P_{L-1}^n, P_0^n}\right|\right| & \cdots & \left|\left|\overrightarrow{P_{L-1}^n, P_{L-2}^n}\right|\right| \end{bmatrix} \quad (1)$$

where $\left|\left|\overrightarrow{P_i^n, P_j^n}\right|\right|$ denotes the Euclidean distance between P_i^n and P_j^n.

Given the length discrepancies observed in the hand bones of different users, it is imperative to standardize the distance features. The key point estimation algorithm, which exhibits superior performance in predicting the metacarpal bones, and the observed stability in the lengths across different postures, suggest that the average lengths of the four metacarpal bones serve as a suitable benchmark value for normalizing the distance features.

Finally, the normalized relative distance feature $\hat{F}_{local_dis}^n$ is flattened into a one-dimensional vector as an input to the network model. This vector has a feature dimension of $\frac{L(L-1)}{2}$.

- **Bending angle feature**

The bending angle feature primarily describes gestures such as grabbing and reflects the bending degree of each finger. We compute the joint angles of each finger independently, as illustrated in Figure 1. Taking the thumb as an example, defining the set of finger joints $S = \{P_1, P_2, P_3, P_4\}$, and adding the wrist key point P_0. The three joints are extracted from S in the order of the joints, e.g. P_0, P_1, P_2. The angle between $\overrightarrow{P_1P_0}$ and $\overrightarrow{P_1P_2}$, denoted a_{0-1-2}, is calculated as a feature describing the flexion of the finger. Similarly, all inter-articular pinch angles were calculated and concatenated into one-dimensional vectors as inputs to the network model, which has a feature dimension number of 50.

- **Translation change feature**

The translation change feature is employed to describe gestures such as hand waving, left and right sliding, and so forth. The local features of these gestures exhibit minimal variation throughout the entirety of the gesture sequence. Consequently, the local gesture features are unable to adequately represent the overall change in the coordinates of the key points. In order to address this issue, we calculate the coordinate change vectors between adjacent frames in order to obtain the translation change features. The calculation

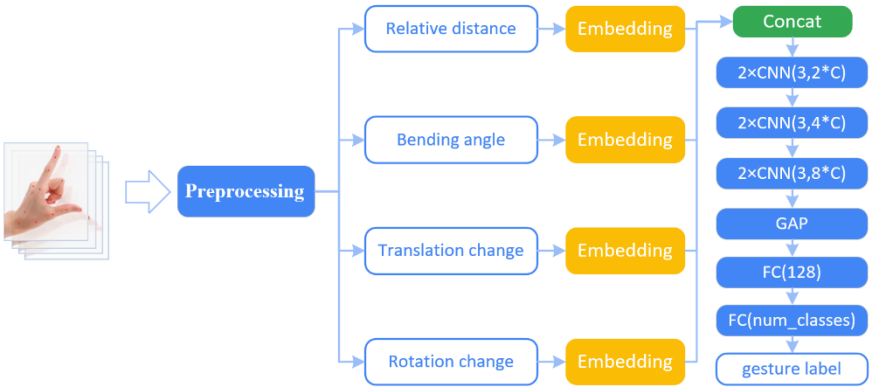


FIGURE 2. The lightweight multi-feature fusion recognition network architecture.

formula is as follows:

$$F^n_{global_{trans}} = S^{n+1} - S^n, \quad n \in \{1, 2, 3, \ldots, N-1\}; \quad (2)$$

The obtained translation change feature $F^n_{global_trans}$ has a length of $N-1$. To keep consistent with the aforementioned local features in the time dimension, $F^n_{global_trans}$ is linearly interpolated to N frames as the input to the network model.

- **Rotation change feature**

The rotational change feature is employed to describe hand gestures such as clockwise rotation and counterclockwise rotation, serving as a supplement to the translation change feature. Specifically, the rotation angle of each keypoint around the palm center is calculated to obtain the overall rotational change feature. The calculation formula is as follows:

$$F^n_{global_{rotate}} = Angle\left(\overrightarrow{P^n_{palm}P^n_l}, \overrightarrow{P^{n+1}_{palm}P^{n+1}_l}\right), \quad n \in \{1, 2, 3, \ldots, N-1\}, \quad l \in \{1, 2, 3, \ldots, L-1\}; \quad (3)$$

where $Angle()$ represents the calculation of the angle between two vectors. Similar to the translation change dimension, a linear interpolation of $F^n_{global_rotate}$ is required, and then used as input to the network model.

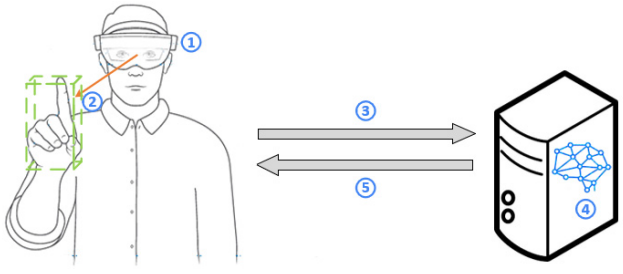


FIGURE 3. The gesture recognition overall framework.

2) NETWORK ARCHITECTURE

This paper uses the DD-Net model framework [18] to fuse multi-dimensional skeleton features. The overall architecture is shown in Figure 2. Through multi-dimensional feature pre-processing on the hand keypoint sequences, relative distance features $F_{local_dis} \in \mathbb{R}^{N\times(L(L-1)/2)}$, bending angle features $F_{local_ang} \in \mathbb{R}^{N\times 50}$, translation change features $F_{global_trans} \in \mathbb{R}^{N\times 3L}$, and rotation change features $F_{global_rotate} \in \mathbb{R}^{N\times(L-1)}$ are obtained. Each pre-processed feature is fed into an embedding layer to obtain embedded features, which aims to extract local relational information within the features and mitigate the influence of skeletal noise. Specifically, the feature embedding layer is defined as

$$Conv1D\left(1, 2^*channels\right) \rightarrow Conv1D\left(3, channels\right) \rightarrow Conv1D(1, channels) \rightarrow Maxpooling(2)$$

where $Conv1D(k, c)$ represents a 1D convolutional layer with kernel size k and c channels. $channels$ is a hyperparameter used to adjust the model capacity.

Then, the embedded features are concatenated as follows:

$$f = f_{local_{dis}} \oplus f_{local_{ang}} \oplus f_{global_{trans}} \oplus f_{global_{rotate}}, \quad f \in \mathbb{R}^{\left(\frac{N}{2}\right)\times(4\times channels)}; \quad (4)$$

where $\oplus$ denotes concatenation operation along the feature dimension.

Subsequently, the concatenated embedded features f are fused through multiple 1D convolutional layers to model global and temporal information. Finally, a classification head composed of fully connected layers predicts the dynamic gesture label.

B. REAL-TIME GAZE-GUIDED RECOGNITION TRIGGER MECHANISM

In mixed reality human-computer interaction scenarios, gaze data is often used as a modality to select interaction targets. The gaze modality responds very quickly, with eye rotation speeds exceeding $300°/s$ for certain movements [19]. Furthermore, fixating on targets of interest is a natural human behavior that does not impose additional interaction overload on users.

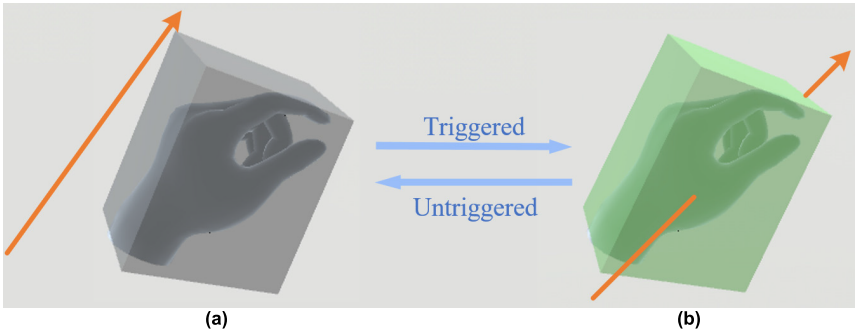


FIGURE 4. The bounding box visualization.

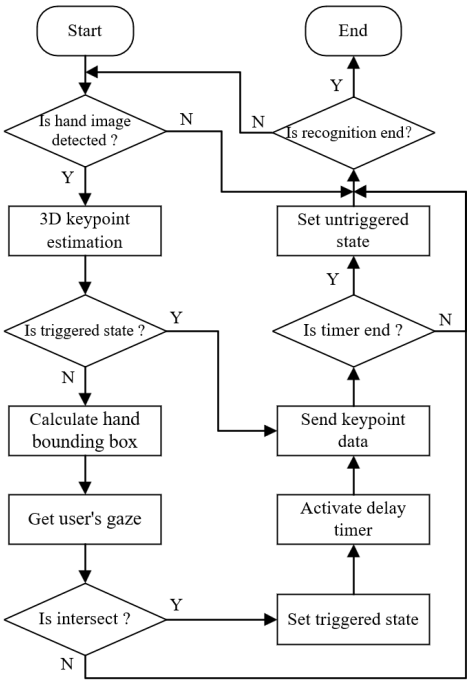


FIGURE 5. The gaze-guided recognition trigger mechanism workflow.

Therefore, this paper uses gaze tracking data to guide the recognition process by naturally and efficiently segmenting the temporal windows of dynamic gestures. The overall workflow is illustrated in Figure 3 and consists of the following steps:

① HoloLens 2 captures hand images and estimates 3D coordinates of hand keypoints and hand bounding box.
② The recognition algorithm is triggered when the user's gaze intersects with the bounding box.
③ Hand keypoint coordinate data is sent from HoloLens 2 to an external PC device.
④ The dynamic gesture recognition algorithm calculates gesture types.
⑤ Gesture types are returned to HoloLens 2.

To avoid ethical and privacy concerns, gaze tracking data is captured and processed locally on the HoloLens 2 device, ensuring that no private personal data is leaked.

Specifically, the triggering process is as shown in Figure 5. HoloLens 2 continuously captures hand images and performs 3D keypoint estimation. Only when the gaze guidance is in the triggered state, the keypoint data will be sent to the external recognition program. The trigger condition is the intersection between the user's gaze and the scaled hand bounding box, where the scaling ratio k is used to set the trigger sensitivity. After the user triggers gesture recognition, a delay timer is activated, and the hand keypoint data within the subsequent period will be sent to the recognition program.

To improve the accuracy of the user's gesture interaction, we visualize the hand bounding box in a semi-transparent manner. When the user's gaze does not intersect with the bounding box, the bounding box appears semi-transparent gray, as shown in Figure 4(a). At this point, the recognition algorithm is in an untriggered state, and the hand keypoint data will not be sent to the external PC device. When the user wants to make dynamic gestures and looks at the hand, the gaze intersects with the bounding box. The bounding box will then appear semi-transparent green for a period of time afterwards, as shown in Figure 4(b), to indicate that the recognition algorithm is now in a triggered state. During this time, the keypoint data is sent to the external PC during this period. After t time, the bounding box returns to state (a).

The addition of this triggering mechanism only slightly increases the difficulty of user interaction, but significantly reduces the false trigger rate of dynamic gestures and improves the accuracy of time window segmentation.

IV. DATASETS AND EXPERIMENTS

A. DATASETS AND IMPLEMENTATION DETAILS

1) DHG-14/28 DATASET

The DHG-14/28(Dynamic Hand Gesture) dataset [12] contains 14 gesture classes, as shown in Table 1. Each gesture class has two forms:

① using only 1 finger to perform the action.
② using the whole hand to perform the action.

Therefore, the dataset contains two benchmark tasks: 14-class coarse-grained gesture classification and 28-class fine-grained gesture classification. The dataset was recorded by 19 volunteers who performed each gesture 5 times,

resulting in a total of 2,800 gesture sequences with lengths ranging from 20 to 50 frames. As shown in Figure 6, each frame contains a depth image and 22 hand keypoint coordinates: 1 palm center (Palm), 1 wrist position (Wrist), and each finger has 4 keypoints, representing fingertip (Tip), 2 articulations and the position connected to the palm (Base). All data was collected by Intel RealSense cameras.

TABLE 1. List of the gestures included in the DHG-14/28 dataset.

Gesture	Label	Tag name
Grab	Fine	G
Expand	Fine	E
Pinch	Fine	P
Rotation CW	Fine	R-CW
Rotation CCW	Fine	R-CCW
Tap	Coarse	T
Swipe Right	Coarse	S-R
Swipe Left	Coarse	S-L
Swipe Up	Coarse	S-U
Swipe Down	Coarse	S-D
Swipe X	Coarse	S-X
Swipe V	Coarse	S-V
Swipe +	Coarse	S-+
Shake	Coarse	Sh

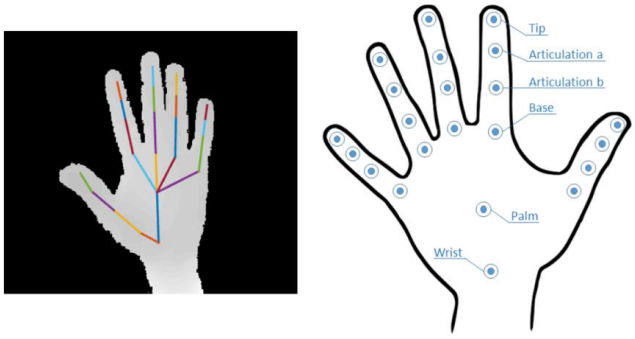


FIGURE 6. Depth and hand skeleton of the DHG-14/28 dataset.

2) SHREC'17 TRACK DATASET

The SHREC'17 Track Dataset [13] is basically the same as the DHG-14/28 Dataset, containing 14 gesture classes. Each gesture class has two forms: (1) using only 1 finger to perform the action, (2) using the whole hand to perform the action. Each gesture was recorded 1-10 times by 28 volunteers. The dataset contains a total of 2,800 gesture sequences, of which 1,960 (70%) are for training and 840 (30%) are for testing. The length of the gesture sequences is 20-50 frames. Each frame contains a depth image and coordinates of 22 hand keypoints. All data was collected by Intel RealSense cameras.

3) HOLOGESTURE DATASET

A visual analysis of the 3D skeleton data from the DHG-14/28 dataset and the SHREC'17 Track dataset was performed, as shown in Figure 7. It can be seen that the hand keypoint data collected by the Intel RealSense camera has large errors in some cases, especially when the hand moves quickly or is perpendicular to the camera. In these cases, the hand pose can hardly be represented correctly. These erroneous data severely affect the algorithm's ability to perform gesture recognition.



FIGURE 7. Visualization of Erroneous data in the DHG-14/28 and SHREC'17 Track datasets.

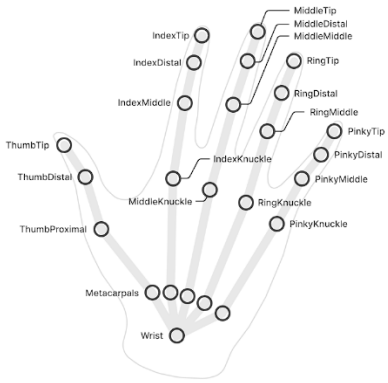


FIGURE 8. Hand tracking skeleton structure on HoloLens 2.

In addition, the hand skeleton structures in the above datasets are different from those of HoloLens 2. They cannot be directly applied to mixed reality scenarios with HoloLens 2, whose hand skeleton structure is shown in Figure 8. Therefore, we use HoloLens 2 to re-record 14 gesture classes, with 2 forms per class, consistent with the above datasets. Each gesture has 150 samples, for a total of 4,200 gesture sequences, of which 2,800 are used for training and 1,400 for testing.

4) IMPLEMENTATION DETAILS

All training and testing in this paper was performed on an RTX 4060 Laptop GPU using CUDA 11.7. Due to the small amount of hand skeleton sample data, each training batch was set to have 512 samples. The default setting for *channels* is 16. The Adam optimizer [26] was used with a learning rate of 1×10^{-3}, which slowly decreased to 1×10^{-5}. During training, if the test accuracy did not improve after 50 epochs, the training was terminated early. For real-time gesture recognition scenarios, the scaling ratio of the triggering bounding box $k = 0.8$, and the timer window size $t = 1.2s$. All

experiments are carried out under the framework of PyTorch 2.0.1 on Windows 11.

TABLE 2. Comparison with other methods on the DHG-14/28 dataset.

Method	14 gestures (%)	28 gestures (%)
SoCJ+HoHD+HoWR [12]	83.1	80.0
CNN+LSTM [20]	85.6	81.1
ST-GCN [21]	91.2	87.1
DG-STA [22]	91.9	87.1
HPEV [23]	92.5	88.9
DSTA-Net [24]	**93.8**	90.9
gVar-FL-Fusion [41]	90.2	88.6
MSFM [42]	87.1	84.4
Graph-General-DNN [36]	92.0	88.8
Ours	92.5	**91.2**

B. ISOLATED GESTURE RECOGNITION EXPERIMENTS

1) RESULTS ANALYSIS ON THE DHG-14/28 DATASET

The lightweight multi-feature fusion recognition network proposed in this paper was trained and tested on the DHG-14/28 dataset. The test results are shown in Table 2.

From Table 2, is can be seen that the proposed method performs well on both the 14-class and 28-class dynamic gesture recognition tasks. The SoCJ+HoHD+HoWR method by De Smedt et al. [12] extracts features using three feature description methods: SoCJ (Shape of Connected Joints), HoHD (Histogram of Hand Directions) and HoWR (Histogram of Wrist Rotations). The hand-crafted features can effectively recognize some gestures with high discriminability, but they cannot fully exploit the implicit features in the data, and thus have lower recognition accuracy compared to other neural network based methods. Nunez et al. [20] adopted a deep learning method combining CNN and LSTM, and experimentally validated that the CNN model architecture can extract relevant features from 3D skeleton data for dynamic recognition using LSTM. Both ST-GCN [21] and DG-STA [22] employ graph convolutional neural network architectures for skeleton-based dynamic gesture recognition. Such architectures can learn the connections between skeletal nodes and better utilize the skeleton graph as prior knowledge, thus achieving higher recognition accuracy. Although the method proposed in this paper does not directly utilize skeleton graph information through GCN, it performs multi-dimensional feature preprocessing on the original skeleton data, extracting skeletal information based on various prior knowledge. This results in features that are more conducive to recognition, and also achieves high recognition accuracy. Liu et al. [23] decoupled gestures into hand pose variation and hand motion variation, and modeled them separately using convolutional networks. We aggregate the hand variations in multiple dimensions by embedding layers for unified modeling. The accuracy is close to the HPEV method in 14-class gesture recognition, and is 2.3% higher in the 28-class recognition task. Shi et al. [24] utilized self-attention mechanisms to model the spatial-temporal dependencies between joints without knowing the connections. It achieved high recognition accuracy in the 14-class task but decreased by 2.8% in the 28-class task, showing limitations in recognizing more gesture classes. gVar-FL-Fusion [41] employed a multimodal feature-level fusion CRNN for dynamic hand gesture recognition, utilizing quantized depth values and 2D skeletal coordinates. MSFM [42] introduced a novel method for encoding spatio-temporal information to enhance the key features necessary for gesture recognition; however, the recognition accuracy of this method remains suboptimal. Graph-General-DNN [36] proposed a multi-branch attention-based graph and a general deep learning model to recognize hand gestures by extracting various types of skeleton-based features. While the model achieved high accuracy on a 14-class gesture recognition task, its performance significantly declined when the number of gesture types increased to 28. The method proposed in this paper achieves a 91.2% recognition accuracy in the 28-class gesture recognition task, surpassing all other methods in the comparison table. However, for the 14-class recognition task, the accuracy is 92.5%, which is 1.3% lower than that of DSTA-Net. We believe this discrepancy is due to the DHG-14 dataset labeling two distinct action forms as the same gesture type, along with key point labeling errors. These issues hinder the model's ability to perform effective feature extraction, resulting in confusion and misclassification.

TABLE 3. Comparison with other methods on the SHREC'17 Track dataset.

Method	14 gestures (%)	28 gestures (%)
Key-frame CNN [13]	82.9	71.9
CNN+LSTM [20]	89.8	86.3
Parallel CNN [25]	91.3	84.4
ST-GCN [21]	92.7	87.7
DG-STA [22]	94.4	90.7
DD-Net [18]	94.6	91.9
HPEV[23]	94.9	92.3
TCN-Summ [37]	93.6	91.4
A-ResLSTM [39]	95.3	91.6
MMEGRN [38]	92.2	89.5
STr-GCN [40]	93.4	89.2
Ours	**95.6**	**92.4**

2) RESULTS ANALYSIS ON THE SHREC'17 TRACK DATASET

The proposed method was also trained and tested on the SHREC'17 Track dataset. The test results are shown in Table 3. The proposed method achieves the best performance on both the 14-class and 28-class dynamic gesture recognition tasks. The Key-frame CNN method by De Smedt et al. [13] performs gesture recognition by applying convolutional neural networks to fused key frame depth images. Since it does not use hand skeleton data, which better describes

poses, its recognition performance is inferior. Devineau and Guillaume [25] proposed a method using parallel convolution to process hand keypoint sequences, proving the performance advantage of convolutional networks in processing hand skeleton data. DD-Net [18] uses a method similar to ours, but it only uses keypoint distance features and frame-to-frame coordinate change features for gesture recognition, without fully exploiting the data information. Its accuracy is 1% and 0.5% lower than our proposed method on the 14-class and 28-class tasks respectively. A-ResLSTM [39] introduced a key gesture spotting architecture, which consists of a novel gesture classifier model and a single-time activation algorithm. Str-GCN [40] separated the spatial and temporal learning of gestures by utilizing Graph Convolutional Networks (GCNs) in combination with Transformers.

TABLE 4. Results on the HoloGesture dataset.

Channels	Parameters (M)	14 Gestures (%)	28 Gestures (%)
2	0.02	90.29	69.79
4	0.03	98.36	96.64
8	0.06	99.29	98.79
16	0.15	99.43	99.36
32	0.50	99.50	99.43
64	1.80	99.43	99.50

3) RESULTS ANALYSIS ON THE HOLOGESTURE DATASET

The lightweight multi-feature fusion recognition network was also trained and tested on the HoloGesture dataset to validate its recognition performance when noise and erroneous data are minimal. The test results are shown in Table 4. When the dataset quality is high, the proposed method can achieve very high recognition accuracy on both 14-class and 28-class dynamic gesture recognition tasks, which meets the requirements of real-time gesture interaction in mixed reality scenarios.

By setting different numbers of *channels*, the model size can be adjusted. As the model size increases, the recognition accuracy steadily increases as well. We choose *channels* = 16 as the recognition model for the following experiments. Under this setting, model size and recognition accuracy reach a balanced point.

4) ABLATION STUDY

To verify the impact of multiple features on gesture recognition performance, we conducted ablation experiments on SHREC'17 Track dataset. The experimental results are presented in Table 5, with the values in parentheses indicating the relative changes. First, we define the network utilizing all features as the baseline. We then sequentially remove each feature class to evaluate their individual contributions. Lastly, we compare models that use only local features or only global features.

The experimental results demonstrate that all four feature types (relative distance, bending angle, translation change, and rotation change) contribute to the gesture recognition task with varying levels of importance. Notably, the removal of translation change results in a significant drop in model performance. When only relative distance and bending angle features are used, there is a 31.6% relative decrease in performance on the 28-class gesture classification task, indicating that these two feature sets alone are insufficient for effective recognition of complex gestures. Overall, the combined use of all features is essential for achieving highly accurate gesture recognition.

TABLE 5. Ablation study on the SHREC'17 Track dataset.

Feature				SHREC'17 Track Dataset	
Relative distance	Bending angle	Translation change	Rotation change	14 Gestures (%)	28 Gestures (%)
✓	✓	✓	✓	95.6	92.4
	✓	✓	✓	89.9 (-6.0)	89.8 (-6.1)
✓		✓	✓	91.8 (-4.0)	90.8 (-5.0)
✓	✓		✓	78.0 (-18.4)	70.1 (-26.7)
✓	✓	✓		91.8 (-4.0)	91.5 (-4.3)
		✓	✓	92.5 (-3.2)	86.1 (-9.9)
✓	✓			74.9 (-21.7)	65.4 (-31.6)

C. REAL-TIME GESTURE RECOGNITION EXPERIMENTS

1) EVALUATION METRICS

Unlike isolated gesture recognition which only considers recognition accuracy, real-time gesture recognition in practical scenarios must also consider the following situations:

① the recognition algorithm misclassifies gestures;
② the time segmentation algorithm has missed detections or false detections;
③ a single gesture is detected multiple times.

Considering the above situations, we use Levenshtein accuracy [27] as the evaluation metric for real-time gesture recognition. Levenshtein accuracy is obtained by dividing the Levenshtein distance [28] by the length of the true gesture sequence. The calculation formula is as follows:

$$L_{acc}\left(1-\frac{L_{dis}}{S}\right)\times 100 \tag{5}$$

where S is the number of gestures in the ground truth sequence, and L_{dis} is the Levenshtein distance, which measures the dissimilarity between two sequences by counting the number of item-level changes (insertion, deletion, or substitutions) required to transform one sequence into the other. For example, the Levenshtein distance between the sequence [1,2,3,4,5,6,7,8,9] and [1,2,7,4,5,6,6,7,8,9] is 2: by substituting "7" with "3", and deleting the duplicate "6" in the second sequence, it can be transformed into the first sequence. For this example, the Levenshtein accuracy is $(1-2/9)\times 100\% = 77.78\%$.

2) SIMULATED MIXED REALITY EXPERIMENT

To test the real-time gesture interaction performance of the above methods in HoloLens 2 mixed reality scenarios,

TABLE 6. Results in the simulated interaction scene (14 gestures).

id	Number of Triggers	Number of False Triggers	Number of Correct Recognitions	False Trigger Rate (%)	Recognition Accuracy (%)	Levenshtein Accuracy (%)	Average Time (ms)
1	101	1	98	1.0	98.0	97.0	3.66
2	103	3	99	2.9	99.0	96.1	3.83
3	102	2	98	2.0	98.0	96.1	3.67
4	104	4	100	3.8	100.0	96.2	3.91
5	102	2	96	2.0	96.0	94.1	3.75
Mean	102.4	2.4	98.2	2.34	98.2	95.9	3.76

TABLE 7. Results in the simulated interaction scene (28 gestures).

id	Number of Triggers	Number of False Triggers	Number of Correct Recognitions	False Trigger Rate (%)	Recognition Accuracy (%)	Levenshtein Accuracy (%)	Average Time (ms)
1	104	4	98	3.8	98.0	94.2	3.94
2	103	3	98	2.9	98.0	95.1	3.74
3	102	2	97	2.0	97.0	95.1	3.71
4	102	2	99	2.0	99.0	97.1	3.81
5	103	3	94	2.0	94.0	91.3	3.89
Mean	102.8	2.8	97.2	2.54	97.2	94.56	3.82

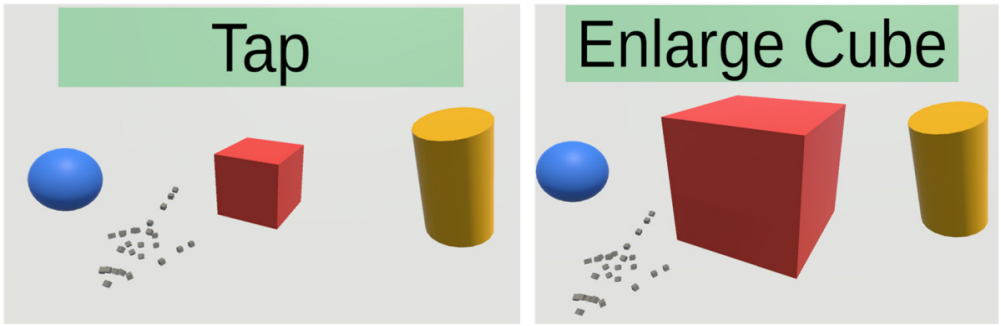


FIGURE 9. Real-time gesture recognition simulated interaction scene.

we designed a simulated interaction scene as shown in Figure 9(a). After training, the testers wore HoloLens 2 and performed 100 randomly prompted dynamic gestures on the screen. To make the interaction scene closer to real use cases, some common interactive actions, such as moving the sphere to the left, enlarging the cube, and rotating the cylinder, are also randomly prompted on the screen, as shown in Figure 9(b). By adding these additional interactive actions, the false trigger rate of the real-time gesture recognition algorithm can be tested. 5 rounds of experiments were conducted in the 14-class and 28-class real-time dynamic gesture recognition scenarios, respectively. In each round, the testers were asked to perform 100 dynamic gestures at random. The experimental results are shown in Table 5 and Table 6. In the 14-class and 28-class gesture recognition tasks, the recognition algorithm was triggered an average of 102.4 and 102.8 times, respectively. The average false trigger times were only 2.4 and 2.8, with false trigger rates of 2.34% and 2.54%. This is mainly due to the introduction of the gaze-guided triggering mechanism, which allows the algorithm to be triggered precisely at the right time.This greatly improves the stability of the gesture interaction. For recognition performance, the accuracy rates were 98.2% and 97.2% for the two tasks, with Levenshtein accuracies of 95.9% and 94.5%, respectively. The average recognition time per gesture was 3.76ms and 3.82ms, respectively, indicating that the recognition network can quickly and accurately recognize user gestures, which is suitable for the mixed reality interaction scenario.

3) STATE-OF-THE-ART COMPARISON

Since the method proposed in this paper introduces gaze tracking data for gesture recognition, there are currently no public datasets that include the gaze modality. Therefore, we list several continuous gesture recognition datasets and mainstream methods, and simply compare their Levenshtein accuracies when performing real-time gesture recognition.

TABLE 8. Comparison with state-of-the-art methods.

Method	Dataset	Input Modality	Levenshtein Accuracy (%)
ResNeXt-101 [29]	IPN-Hand [29]	RGB	25.34
ResNeXt-101 [29]		RGB-Flow	42.47
ResNeXt-101 [29]		RGB-seg	39.01
Resnet-50 [29]		RGB	19.78
Resnet-50 [29]		RGB-Flow	39.47
Resnet-50 [29]		RGB-seg	33.27
TD-Net [30]		Skeleton	40.10
TY-Net [31]		Skeleton	60.9±0.5
ME+EHTD+Bi-LSTM[35]		RGB+Seg+Flow+EHTD	73.71
ResNet-10+ResNeXt-101 [27]	NVGesture [34]	RGB+Depth	77.39
ME+EHTD+Bi-LSTM[35]		RGB+Depth+InfraRed+EHTD	88.32
ResNet-10+ResNeXt-101 [27]	EgoGesture [35]	RGB+Depth	91.04
MotionNet+GestureNet [32]	ZJUGesture[33]	RGB	92.40
Ours	HoloGesture	Skeleton+Gaze	**95.90/94.56**

As shown in Table 7, in IPN-Hand [29] and NVGesture [33] datasets, since the hand is always in the camera view and contains some casual motions unrelated to formal gestures, making it difficult to accurately segment the temporal windows of dynamic gestures, resulting in many misrecognitions and generally low Levenshtein accuracies, making them inapplicable to real interaction scenarios. In the EgoGesture [34] and ZHUGesture [32] datasets, the recorders would move their hands out of the camera view when not performing prescribed dynamic gestures. Therefore ResNet-10+ResNeXt-101 [17] and MotionNet+ GestureNet [32] adopt a strategy of first detecting whether there is a hand image in the view before performing classification and recognition. This strategy is very effective on such datasets, with Levenshtein accuracies above 90%, but in real interaction scenarios, users will make many casual movements irrelevant to dynamic gestures, leading to misrecognition by the algorithms. The method proposed in this paper uses the gaze modality to allow users to actively trigger dynamic gesture recognition, rather than relying on detecting whether the view contains hand images. It is more suitable for real-world interaction scenarios, with Levenshtein accuracies around 95%, able to meet the needs of most interaction scenarios.

V. CONCLUSION

In this paper, we propose a real-time dynamic gesture recognition method based on gaze guidance, which uses gaze tracking data to accurately segment hand skeletal sequences. Meanwhile, the skeletal data is pre-processed with multi-dimensional feature extraction. Finally, a lightweight multi-feature fusion recognition network performs gesture recognition. Experiments on the DHG-14/28 dataset, the SHREC'17 Track dataset, and our self-built HoloGesture dataset validate that with only about 0.15M parameters and 3ms inference time, the proposed method achieves higher recognition accuracy than mainstream methods. We also constructed a simulated mixed reality interaction scenario to evaluate the real-time performance. The results show that in the 14-class and 28-class dynamic gesture recognition tasks, our method achieves Levenshtein accuracies of 95.9% and 94.5% respectively, improving mainstream methods by about 20%. This enables stable, fast, and accurate real-time dynamic gesture recognition.

The methodology presented in this paper holds significant potential for a range of practical applications. For instance, it can be applied in virtual and augmented reality (AR/VR) environments to enable more intuitive control of virtual objects. Additionally, it has potential in healthcare for facilitating touchless interactions in sterile settings, such as surgical rooms. This technology may also benefit the gaming industry by enhancing user experiences through immersive, gesture-based controls.

While the proposed method enables fast and accurate gesture recognition, there are still some limitations in this research. Specifically, the method proposed in this paper requires access to gaze tracking data. Although an increasing number of hardware devices now support eye-tracking, it will take time for this modality to become widely adopted. Additionally, the accuracy of gesture recognition depends on the precision of both hand skeletal and gaze tracking data. If these data are not acquired accurately, the effectiveness of gesture recognition may be compromised.

Although the proposed lightweight multi-feature fusion recognition network and gaze-guided recognition mechanism demonstrate high accuracy and efficiency, future research could explore integrating additional input modalities such as voice commands or haptic feedback to create a more immersive human-computer interaction experience. Another avenue of research could focus on optimizing the system for various hardware platforms, such as low-power wearable devices, to enhance portability and scalability.

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